

JEE MAIN REVISION MASTER: HEIGHT AND DISTANCE OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 40 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Height And Distance — Core Principles & Governing Laws	37 PYQs	Core Principle
Moderate	Height And Distance — Standard Calculations & Problem Archetypes	4 PYQs	Quantitative Analysis
Standard	Height And Distance — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Height And Distance — Core Principles & Governing Laws	$TM = \frac{h}{\tan \theta}$, $\tan(90^\circ - \theta) = \frac{x}{h}$
Height And Distance — Standard Calculations & Problem Archetypes	$LCPA = \frac{h}{\tan \theta}$, $AB = x$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Height And Distance — Core Principles & Governing Laws (Very High Yield — 37 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Height And Distance.

Caution / JEE Pitfall: Carefully verify units, standard reference states, and sign conventions in Height And Distance.

Speed Trick: Focus on high-yield formulas and standard limiting approximations tested in Height And Distance PYQs.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ):** "The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due North of the tower is 30° . The height of the tower is $10\sqrt{3}$ m. Find the distance between the two persons." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ):** "From the top A of a vertical wall AB of height 30m, the angles of depression of the top P and bottom Q of a vertical tower PQ are 30° and 60° respectively. The height of the tower PQ is x m. Find the value of x ." [Ans: N/A]
- [2022 Evening Shift] Q#1 (MCQ):** "The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45° . Let R be a point on the ground such that AR is perpendicular to the line PQ. The angle of elevation of P from R is θ . Find the value of $\tan \theta$." [Ans: N/A]

3.2 Height And Distance — Standard Calculations & Problem Archetypes (Moderate Yield — 4 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Height And Distance.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due West of the tower is 30° . If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to" [Ans: N/A]
- [2021 Morning Shift] Q#2 (MCQ): "A vertical pole fixed to the horizontal ground is divided in the ratio 3 : 7 by a mark on it with lower part shorter than the upper part. If..." [Ans: N/A]
- [2017 Shift] Q#3 (MCQ): "Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB and P be a point on the ground such that $AP = 2AB$" [Ans: N/A]

3.3 Height And Distance — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Height And Distance.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due West of the tower is 30° . If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to" [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "From the top A of a vertical wall AB of height 30m, the angles of depression of the top P and bottom Q of a vertical tower PQ are..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 11th April Evening Shift (Q#1)

The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due West of the tower is 30° . If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to

- (A) 10
- (B) $(5/2)\sqrt{5}$
- (C) $5\sqrt{5}$
- (D) 5

Correct Answer: None

Solution Analysis: Let's denote the person standing due south as S and the one standing due west as W. Also, let the tower be at point T. From person S's perspective, we have a right triangle $\triangle SPT$. The height of the tower PT is given as 5m, which is the opposite side for angle S. The angle at S is 45° . From the tangent trigonometric ratio, we have : $\tan 45^\circ = (PT / ST) = (5 / ST)$ Since $\tan 45^\circ = 1$, we get $ST = 5$ m. Similarly, from person W's perspective, we have a right triangle $\triangle WPT$. The angle at W is 30° . From the tangent trigonometric ratio, we have: $\tan 30^\circ = (PT / WT) = (5 / WT)$

Benchmark Problem 2: JEE Main 2021 (Online) 31st August Morning Shift (Q#2)

A vertical pole fixed to the horizontal ground is divided in the ratio 3 : 7 by a mark on it with lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18 m away from the base of the pole, then the height of the pole (in meters) is :

- (A) $12\sqrt{15}$
- (B) $12\sqrt{10}$
- (C) $8\sqrt{10}$
- (D) $6\sqrt{10}$

Correct Answer: None

Solution Analysis: Let height of pole = $10l$ $\tan\alpha = (3l / 18) = (l / 6)$ $\tan 2\alpha = (10l / 18)$ $\frac{2\tan\alpha}{1-\tan^2\alpha} = (10l / 18)$ use $\tan\alpha = (l / 6) \Rightarrow l = \sqrt{(72 / 5)}$ height of pole = $10l = 12\sqrt{10}$